Six-sided die, marked in the usual way, 1 thru 6

It is <u>not</u> fair, but know odd and even have equal prob. ($\frac{1}{2}$)

X = outcome of toss

$\Omega = \{1,2,3,4,5,6\}$

$\mathcal{F} = \{\{1,3,5\}, \{2,4,6\}, \emptyset, \Omega\}$

So, $P(X \le 2) = P(\underbrace{\{\omega : X(\omega) \le 2\}}_{\notin \mathcal{F}})$ is undefined.

Exercise: Suppose that random variable X has pmf

$$f_X(k) = \frac{1}{2k^2 z(2)}, \quad k = \pm 1, \pm 2, \ldots$$

compare with Zeta dist. with $s=2$

What is $E(X)$ in this case?

Technical assumption: In order for $E(X)$ to exist,

$$E(X) = \underbrace{\sum_{\substack{k \geq 0: \\ f_X(k) > 0}} k f_X(k)}_{1} + \underbrace{\sum_{\substack{k < 0: \\ f_X(k) > 0}} k f_X(k)}_{2}$$

need either 1 is finite or 2 is finite.

I.e., Cannot end up with $\infty - \infty$.

In this example, $E(X)$ "does not exist"

$$\cancel{E(X+Y) = E(X) + E(Y)}$$

$$\cancel{Y = -X \qquad E(X) = \infty}$$

$$E(X) = \sum_{k: f_X(k) > 0} k f_X(k) = \underbrace{(1) f_X(1) + (-1) f_X(-1) + \cdots}_{0}$$

$$1 + 2 + 3 + \cdots = -\frac{1}{12}$$

Rule for the Lazy Statistician

It is possible to prove that

$$E(h(X)) = \sum_{k:\, f_X(k) > 0} h(k) f_X(k)$$

This result, sometimes called **the Rule for the Lazy Statistician**, is a **theorem**, and not a **definition**.

Exercise: What would the **definition** of $E(h(X))$ be?

Let $Y = h(X)$, i.e. $Y(\omega) = h(X(\omega))$

Derive its pmf f_Y, then

$$E(Y) = E(h(X)) = \sum_{k:\, f_Y(k)>0} k f_Y(k)$$

Rule for Lazy Stat says don't need to derive f_Y, can be "lazy"

An important case is the **variance** of a random variable, defined to be

$$V(X) = E\left((X - E(X))^2\right) = E(X^2) - E(X)^2.$$

The **standard deviation** is the square root of the variance.

"Named" Discrete Distributions

Some distributions are so widely utilized that they have "names." Keep in mind, however, that these are simply special cases of the general class of discrete distributions.

The distributions we will consider here are as follows:

1. The Bernoulli(p) Distribution
2. The Binomial(n, p) Distribution
3. The Geometric(p) Distribution
4. The Poisson(λ) Distribution

In each case, there are **parameters** that specify the exact form of the distribution. So, in fact, there is not "**a** Binomial Distribution," but instead a "**family** of Binomial Distributions."

The Bernoulli(p) Distribution

A **Bernoulli Trial** is an experiment that has two possible outcomes (typically called "success" and "failure.")

X is said to have the **Bernoulli(p) distribution** if $X = 1$ when the trial is a "success" and $X = 0$ when it is a "failure," and $P(X = 1) = p$.

Exercise: What is the pmf of X in this case? What are $E(X)$ and $V(X)$?

$$f_X(k) = \begin{cases} p, & k=1 \\ 1-p, & k=0 \end{cases}$$

$$E(X) = p$$

$$V(X) = E(X^2) - E(X)^2$$
$$= p - p^2$$
$$= p(1-p)$$

The Binomial(n, p) Distribution

Consider a sequence of n independent Bernoulli trials, and for each trial $P(\text{success}) = p$. If X is the number of "successes," in those n trials, then X has the **binomial(n, p) distribution**.

Example: If I roll a die ten times, and X equals the number of "1"'s, then we write $X \sim \text{binomial}(n = 10, p = 1/6)$. In this case, "success" is defined as "rolling a '1'."

If $X \sim \text{binomial}(n, p)$, then

$$f_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

and $E(X) = np$ and $V(X) = np(1-p)$.

Exercise: A life insurance company is interested in the number of claims that it will receive on a particular block of life insurance policies in the coming year. All the policies in the block are written on lives of people who are currently age 65. There are presently 100 active policies in the block. (None of the policies in the block are written on more than one life, and no insured person is covered by more than one policy.). Based on the United States Life Tables the company estimates that the probability of any given 65-year-old dying in the next year is 0.0145 (In actuarial notations, $q_{65} = 0.0145$.). Assume that the future lifetime of the insured persons are independent.

Determine the probability that the company receives more than two claims on this block of 100 policies in the coming year.

$X = \#$ of claims

$X \sim \text{binomial}(n = 100,\ p = 0.0145)$

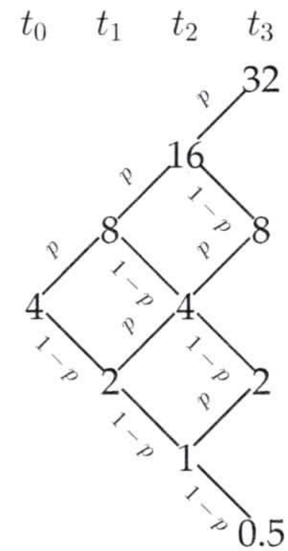
$P(X > 2) = 1 - F_X(2)$

In Python: import scipy.stats as sc

$1 - \text{sc.binom.cdf}(2, \underset{n}{100}, \underset{p}{0.0145})$

$= 0.178$

Exercise: The **binomial model** for stock prices describes the evolution of prices in discrete time. Starting at a value S, the value in the next period could be uS with probability p or dS with probability $1-p$. Assume that $u > d$. It is often assumed that $u = 1/d$, in which case a simple tree can be used to illustrate the possible paths. For example, if $S_0 = 4$, $u = 2$, and $d = 1/2$, then the model has the following structure:



Let S_t denote the value at time t. Describe the distribution of S_t.

Assume start at $t=0$, future times are
$t=1,2,\dots$

Then S_t has $t+1$ possible values
so, S_t is a discrete RV

The possible values for S_t can be written as $S_0 u^k d^{t-k}$, $k = 0, 1, \ldots, t$

where k = # of "up" steps

$$P(S_t = S_0 u^k d^{t-k}) = P(\text{exactly } k \text{ up steps})$$

$$= \binom{t}{k} p^k (1-p)^{t-k} \quad k = 0, 1, \ldots, t$$

since # of up steps has binomial(t, p) distribution

Note: <u>Assuming</u> steps are independent.

S_t does <u>not</u> have a binomial dist.

Comment: $S_t = S_0 X_1 X_2 X_3 \cdots X_t$ where $X_1, X_2, \ldots, X_t$ are independent RVs each is such that $P(X_i = u) = p \quad P(X_i = d) = 1 - p$

So, $$\log S_t = \log S_0 + \underbrace{\sum_{i=1}^{t} \log X_i}_{\text{sum of independent, identically dist. RVs}}$$

As $t \nearrow \infty$, CLT says that $\log S_t \approx$ normal

From Stochastic Calculus for Finance I, Sheve

2.2 Random Variables, Distributions, and Expectations

A random experiment generally generates numerical data. This gives rise to the concept of a random variable.

Definition 2.2.1. *Let $(\Omega, \mathbb{P})$ be a finite probability space. A* random variable *is a real-valued function defined on Ω. (We sometimes also permit a random variable to take the values $+\infty$ and $-\infty$.)*

Example 2.2.2 (Stock prices). Recall the space Ω of three independent coin-tosses (2.1.1). As in Figure 1.2.2 of Chapter 1, let us define stock prices by the formulas

$$S_0(\omega_1\omega_2\omega_3) = 4 \text{ for all } \omega_1\omega_2\omega_3 \in \Omega_3,$$

$$S_1(\omega_1\omega_2\omega_3) = \begin{cases} 8 & \text{if } \omega_1 = H, \\ 2 & \text{if } \omega_1 = T, \end{cases}$$

$$S_2(\omega_1\omega_2\omega_3) = \begin{cases} 16 & \text{if } \omega_1 = \omega_2 = H, \\ 4 & \text{if } \omega_1 \neq \omega_2, \\ 1 & \text{if } \omega_1 = \omega_2 = T, \end{cases}$$

$$S_3(\omega_1\omega_2\omega_3) = \begin{cases} 32 & \text{if } \omega_1 = \omega_2 = \omega_3 = H, \\ 8 & \text{if there are two heads and one tail}, \\ 2 & \text{if there is one head and two tails}, \\ .50 & \text{if } \omega_1 = \omega_2 = \omega_3 = T. \end{cases}$$

Here we have written the arguments of S_0, S_1, S_2, and S_3 as $\omega_1\omega_2\omega_3$, even though some of these random variables do not depend on all the coin tosses. In particular, S_0 is actually not random because it takes the value 4, regardless of how the coin tosses turn out; such a random variable is sometimes called a *degenerate random variable*. □

It is customary to write the argument of random variables as ω, even when ω is a sequence such as $\omega = \omega_1\omega_2\omega_3$. We shall use these two notations interchangeably. It is even more common to write random variables without any arguments; we shall switch to that practice presently, writing S_3, for example, rather than $S_3(\omega_1\omega_2\omega_3)$ or $S_3(\omega)$.

According to Definition 2.2.1, a random variable is a function that maps a sample space Ω to the real numbers. The *distribution* of a random variable is a specification of the probabilities that the random variable takes various values. *A random variable is not a distribution*, and *a distribution is not a random variable*. This is an important point when we later switch between the actual probability measure, which one would estimate from historical data, and the risk-neutral probability measure. The change of measure will change distributions of random variables but not the random variables themselves. We make this distinction clear with the following example.

Example 2.2.3. Toss a coin three times, so the set of possible outcomes is

$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Define the random variables

$$X = \text{Total number of heads}, \qquad Y = \text{Total number of tails}.$$

In symbols,

$$\begin{aligned} X(HHH) &= 3, \\ X(HHT) = X(HTH) = X(THH) &= 2, \\ X(HTT) = X(THT) = X(TTH) &= 1, \\ X(TTT) &= 0, \end{aligned}$$

$$\begin{aligned} Y(TTT) &= 3, \\ Y(TTH) = Y(THT) = Y(HTT) &= 2, \\ Y(THH) = Y(HTH) = Y(HHT) &= 1, \\ Y(HHH) &= 0. \end{aligned}$$

We do not need to know probabilities of various outcomes in order to specify these random variables. However, once we specify a probability measure on Ω, we can determine the distributions of X and Y. For example, if we specify the probability measure $\widetilde{\mathbb{P}}$ under which the probability of head on each toss is $\frac{1}{2}$ and the probability of each element in Ω is $\frac{1}{8}$, then

$$\begin{aligned} \widetilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 0\} &= \widetilde{\mathbb{P}}\{TTT\} = \frac{1}{8}, \\ \widetilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 1\} &= \widetilde{\mathbb{P}}\{HTT, THT, TTH\} = \frac{3}{8}, \\ \widetilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 2\} &= \widetilde{\mathbb{P}}\{HHT, HTH, THH\} = \frac{3}{8}, \\ \widetilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = 3\} &= \widetilde{\mathbb{P}}\{HHH\} = \frac{1}{8}. \end{aligned}$$

We shorten the cumbersome notation $\widetilde{\mathbb{P}}\{\omega \in \Omega; X(\omega) = j\}$ to simply $\widetilde{\mathbb{P}}\{X = j\}$. It is helpful to remember, however, that the notation $\widetilde{\mathbb{P}}\{X = j\}$ refers to the probability of a subset of Ω, the set of elements ω for which $X(\omega) = j$. Under $\widetilde{\mathbb{P}}$, the probability that X takes the four values 0, 1, 2, and 3 are

$$\begin{aligned} \widetilde{\mathbb{P}}\{X = 0\} &= \frac{1}{8}, \ \widetilde{\mathbb{P}}\{X = 1\} = \frac{3}{8}, \\ \widetilde{\mathbb{P}}\{X = 2\} &= \frac{3}{8}, \ \widetilde{\mathbb{P}}\{X = 3\} = \frac{1}{8}. \end{aligned}$$

This table of probabilities where X takes its various values records the *distribution* of X under $\widetilde{\mathbb{P}}$.

The random variable Y is different from X because it counts tails rather than heads. However, under $\widetilde{\mathbb{P}}$, the distribution of Y is the same as the distribution of X:

$$\widetilde{\mathbb{P}}\{Y=0\} = \frac{1}{8},\ \widetilde{\mathbb{P}}\{Y=1\} = \frac{3}{8},$$
$$\widetilde{\mathbb{P}}\{Y=2\} = \frac{3}{8},\ \widetilde{\mathbb{P}}\{Y=3\} = \frac{1}{8}.$$

The point here is that the random variable is a function defined on Ω, whereas its distribution is a tabulation of probabilities that the random variable takes various values. A random variable is not a distribution.

Suppose, moreover, that we choose a probability measure $\mathbb{P}$ for Ω that corresponds to a $\frac{2}{3}$ probability of head on each toss and a $\frac{1}{3}$ probability of tail. Then

$$\mathbb{P}\{X=0\} = \frac{1}{27},\ \mathbb{P}\{X=1\} = \frac{6}{27},$$
$$\mathbb{P}\{X=2\} = \frac{12}{27},\ \mathbb{P}\{X=3\} = \frac{8}{27}.$$

The random variable X has a different distribution under $\mathbb{P}$ than under $\widetilde{\mathbb{P}}$. It is the same random variable, counting the total number of heads, regardless of the probability measure used to determine its distribution. This is the situation we encounter later when we consider an asset price under both the actual and the risk-neutral probability measures.

Incidentally, although they have the same distribution under $\widetilde{\mathbb{P}}$, the random variables X and Y have different distributions under $\mathbb{P}$. Indeed,

$$\mathbb{P}\{Y=0\} = \frac{8}{27},\quad \mathbb{P}\{Y=1\} = \frac{12}{27},$$
$$\mathbb{P}\{Y=2\} = \frac{6}{27},\quad \mathbb{P}\{Y=3\} = \frac{1}{27}.$$ □

Definition 2.2.4. *Let X be a random variable defined on a finite probability space $(\Omega, \mathbb{P})$. The* expectation *(or* expected value*) of X is defined to be*

$$\mathbb{E}X = \sum_{\omega \in \Omega} X(\omega)\mathbb{P}(\omega).$$

When we compute the expectation using the risk-neutral probability measure $\widetilde{\mathbb{P}}$, we use the notation

$$\widetilde{\mathbb{E}}X = \sum_{\omega \in \Omega} X(\omega)\widetilde{\mathbb{P}}(\omega).$$

The variance *of X is*

$$\operatorname{Var}(X) = \mathbb{E}\Big[(X - \mathbb{E}X^2\Big].$$

Geometric(p) Distribution

Again, consider a sequence of independent Bernoulli trials, each with probability of "success" equal to p. Let X denote the number of trials **prior to** the first "success." Then $X \sim$ geometric(p).

In other words, X is the number of "failures" prior to the first success.[1]

The pmf in this case is

$$f_X(k) = (1-p)^k p, \quad k = 0, 1, 2, \ldots.$$

Also, $E(X) = (1-p)/p$ and $V(X) = (1-p)/p^2$.

Note: Remember that when $|r| < 1$,

$$\sum_{k=a}^{\infty} r^k = \frac{r^a}{1-r}.$$

[1]Note that in some texts this distribution is defined as the number of trials **up to and including** the first success.

Exercise (From HOTS): You have three children, but only one apple. You want to toss a fair coin to determine which child gets the apple. You want each child to be equally likely to get the apple. What is your strategy? What is the expected number of tosses needed to complete this strategy?

Flip twice

HH ⇒ child 1
HT ⇒ " 2
TH ⇒ " 3
TT ⇒ redo

A "trial" is a pair of flips. A trial is a "success" if a child receives the apple, ie. if the flips is <u>not</u> TT

$X = \#$ of failed trials
$X \sim \text{geometric}(p = 3/4)$

Number of flips $= 2X + \underline{\underline{2}}$

$$E(\#\text{ of flips}) = E(2X+2) = 2E(X) + 2$$
$$= 2\left(\frac{1-p}{p}\right) + 2$$
$$= 8/3$$

Exercise: Three people (call them A, B, and C) roll a die in that order. They repeat this order until the first "6," at which time that person wins the game. What is the probability that A wins? What is the probability that B wins?

X = # of tosses before first "6"

$X \sim$ geometric$(p = 1/6)$

$$P(A \text{ wins}) = P(X=0, X=3, X=6, \ldots)$$

$$= \sum_{k=0}^{\infty} P(X=3k)$$

$$= \sum_{k=0}^{\infty} \left(1-\frac{1}{6}\right)^{3k}\left(\frac{1}{6}\right)$$ using geometric pmf

$$= \frac{1}{6}\sum_{k=0}^{\infty}\left(\left(\frac{5}{6}\right)^3\right)^k$$

$$= \left(\frac{1}{6}\right)\left(\frac{1}{1-(5/6)^3}\right)$$ using geom. series result with $a=0, r=(5/6)^3$ (see Math Facts)

$$= \frac{36}{91}$$

$X=7$

$$P(B \text{ wins}) = P(X=1, X=4, \ldots)$$

Or: $P(B \text{ wins}) = P(B \text{ wins} \mid \text{first roll is "6"}) \times P(\text{first roll is "6"})$

$+ P(B \text{ wins} \mid \text{first roll is not "6"})\, P(\text{first not "6"})$

LFTP

$$= (0)\left(\tfrac{1}{6}\right) + \left(\tfrac{36}{91}\right)\left(\tfrac{5}{6}\right) = \tfrac{30}{91}$$

$$P(C \text{ wins}) = \left(\tfrac{5}{6}\right)\left(\tfrac{30}{91}\right)$$

Or: $P(B \text{ wins}) = \tfrac{5}{6} P(A \text{ wins})$

$P(C \text{ wins}) = \tfrac{5}{6} P(B \text{ wins}) = \tfrac{25}{36} P(A \text{ wins})$

$$P(A \text{ wins}) + P(B \text{ wins}) + P(C \text{ wins}) = 1$$

$$P(A \text{ wins}) + \tfrac{5}{6} P(A \text{ wins}) + \tfrac{25}{36} P(A \text{ wins}) = 1$$

$$P(A \text{ wins}) = 36/91$$

$$P(C \text{ wins}) = P(C \text{ wins} \mid \text{either first } \underline{\text{two}} \text{ rolls are "6"})$$
$$\times P(\text{either first } \underline{\text{two}} \text{ rolls are "6"})$$
$$+ P(C \text{ wins} \mid \text{neither first two rolls are "6"})$$
$$P(\text{neither first two rolls are "6"})$$
$$= (0)\left(\frac{11}{36}\right) + \underbrace{\left(\frac{36}{91}\right)\left(\frac{5}{6}\right)}_{P(B \text{ wins})}\left(\frac{5}{6}\right)$$
$$= \frac{5}{6} P(B \text{ wins})$$

Poisson(λ) Distribution

A random variable X has the **Poisson(λ) Distribution**, for $\lambda > 0$, if it has pmf

$$f_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots.$$

This is useful for modelling counts, especially the number of times something happens in a fixed period of time. It holds that $E(X) = \lambda$ and $V(X) = \lambda$.

The Poisson pmf can be derived via the following considerations: Fix some $\lambda > 0$, and let $X \sim \text{binomial}(n, p = \lambda/n)$ and let $Y \sim \text{Poisson}(\lambda)$. Then for all $k = 0, 1, \ldots$,

$$\lim_{n\to\infty} P(X = k) = P(Y = k).$$

Exercise: Divide a fixed time interval into n equally-sized segments. For each of these subintervals, place a "mark" in it with probability λ/n. (Fix $\lambda > 0$, and choose $n > \lambda$.) Let X equal the total number of "marks." What is the distribution of X? What happens as n grows?

X has the binomial$(n, \lambda/n)$ dist.

As $n \nearrow \infty$ dist. of X becomes the Poisson(λ)

$$P(X=k) = \left(\frac{n!}{(n-k)!\,k!}\right)\left(\frac{\lambda}{n}\right)^k \left(1-\frac{\lambda}{n}\right)^{n-k}$$

$$= \left(\frac{\lambda^k}{k!}\right)\left(\frac{n!}{(n-k)!\,n^k}\right)\left(1-\frac{\lambda}{n}\right)^{n}\left(1-\frac{\lambda}{n}\right)^{-k}$$

as $n \nearrow \infty$

$$\downarrow \qquad \downarrow \qquad \downarrow \qquad \downarrow$$

$$\left(\frac{\lambda^k}{k!}\right)\left(1\right)\left(e^{-\lambda}\right)\left(1\right)$$

$$= P(Y=k)$$

Exercise: An auto insurance company has determined that the average number of claims against the comprehensive coverage of a policy is 0.6 per year.

1. What is the probability that a policy holder will file one claim in a year?
2. What is the probability that a policy holder will file more than one claim in a year?

If I'm willing to assume the claims arrive independently, then $X \sim \text{Poisson}(\lambda = 0.6)$ where $X = \#$ of claims in year

(1) $P(X=1) = \dfrac{e^{-\lambda}\lambda^{1}}{1!} = e^{-0.6}(0.6) = 0.33$

(2) $P(X>1) = 1 - P(X=0) - P(X=1)$
$= 0.12$

Independent Discrete Random Variables

The notion of independence extends naturally to discrete random variables. In particular, for a pair of discrete random variables X and Y, they are said to be **independent** if for all a and b,

$$P(X = a \text{ and } Y = b) = P(X = a)P(Y = b).$$

Exercise: Suppose that X has the Poisson(λ_1) distribution, and that Y has the Poisson(λ_2) distribution. Does $X + Y$ have the Poisson distribution?

For any $k = 0, 1, 2, \ldots$

$$P(X+Y=k) = \sum_{i=0}^{k} P(\underbrace{X=i \text{ and } Y=k-i})$$

$$= \sum_{i=0}^{k} P(X=i)\,P(Y=k-i) \quad \text{Assuming } X, Y \text{ independent}$$

$$= \sum_{i=0}^{k} \left(\frac{e^{-\lambda_1}\lambda_1^{i}}{i!}\right)\left(\frac{e^{-\lambda_2}\lambda_2^{k-i}}{(k-i)!}\right) \quad \text{Poisson pmf}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{i=0}^{k} \lambda_1^{i}\lambda_2^{k-i}\frac{k!}{i!(k-i)!}$$

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!}(\lambda_1+\lambda_2)^{k} \quad \text{by binomial theorem}$$

So, $X+Y$ is Poisson$(\lambda_1+\lambda_2)$ if X, Y are independent